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Strength, Jump Height, Landing, and Mobility Metrics Predict High and Low Performers of a Fire and Move Assessment: A Machine Learning Approach

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BASIC SCIENCES

ABSTRACT

MCCARTHY, A., J. T. FULLER, J. A. WILLS, S. CASSIDY, M. LOVALEKAR, B. C. NINDL, and T. L. A. DOYLE. Strength, Jump Height, Landing, and Mobility Metrics Predict High and Low Performers of a Fire and Move Assessment: A Machine Learning Approach. *Med. Sci. Sports Exerc.*, Vol. 57, No. 12, pp. 2754–2763, 2025. **Purpose:** Combat maneuverability is critical for soldier survivability. Military organizations ensure effective combat maneuverability through routine assessments. Advanced statistical analyses may improve combat movement efficiency practices. This study grouped physical qualities (e.g., strength, power, mobility) via an exploratory factor analysis (EFA) and extracted factors to compare high and low performers and develop predictive models. **Methods:** Thirty-four participants completed two sessions assessing physical qualities and combat movement performance. Participants were classified as either "high" or "low" performers (i.e., completed 50 laps of the assessment or completed less than 50 laps, respectively). An EFA was conducted to reduce physical quality dataset dimensions into specific factors. *T*-test and effect size compared factors between high and low performers. Logistic regression, multilayer perceptron, and random forest models were trained and tested to classify performers based on factor values. Feature importance scores determined factors most influential in classifying participants. **Results:** EFA resulted in four factors (81.46% variance explained). Factor 1 represented isometric strength, jumping, and drop landing ability. Factors 2–4 represent isometric strength and rate of force development in the lower and upper body, and overhead squat ability, respectively. All factors significantly differed between groups, with high performers demonstrating higher mean values than low performers ($P < 0.05$). Factor 1 demonstrated a very large effect size ($d = 2.15$), whereas factors 2–4 were moderate–large ($d = 0.72$–0.81). The logistic regression model had 100% accuracy in the testing phase, whereas other models achieved 86%. Factor 1 was the most influential factor across models (approximately six times more than other factors). **Conclusions:** Utilized models show military applicability in classifying high or low performers for combat maneuverability. Physical interventions optimizing factor 1 may enhance combat maneuverability. **Key Words:** CLASSIFICATION MODELS, COMBAT, PHYSICAL INTERVENTIONS, MILITARY

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The ability to move effectively and efficiently in combat is critical for a soldier's survivability (1,2). The threat to survival is at its highest when a soldier moves under fire from one position of cover to another (3). Integral to survivability is decreased movement time between cover positions; thus, reducing exposure to ballistic wounds (3). Combat soldiers must perform these explosive maneuvers while essential combat gear is externally carried and donned, weighing greater than or equal to 23 kg (4). Physical qualities, such as strength and power, are essential to performing explosive maneuvers, which can help to support survival when under fire (5,6). For example, Stein et al. (5) demonstrated that the standing broad jump and critical running velocity can predict 74% of the variance of a combat movement assessment (7). Foulis et al. (8) also demonstrated that multistage fitness scores were predictive of combat maneuverability;

thus, reinforcing the need for physical assessment standards in military organizations.

Many nations like Australia, the United States, and the United Kingdom require active-duty soldiers to pass simulated combat assessments that challenge the physical qualities needed to effectively move between cover positions to verify that soldiers are deployable and combat ready (9–11). Specifically, the Australian Defence Force (ADF) requires soldiers to complete the fire and movement assessment, where soldiers start in a prone position, accelerate to a run, and then decelerate to a kneeling fire position over a 6-m predetermined course, within 5 s (3). To be marked fit for duty, infantry soldiers must complete 16 of these "bounds," followed by an 18-m leopard crawl while wearing a specified combat load (7,10). In addition, Australian Special Forces Units must complete a 1-km loaded run immediately before completing the fire and movement assessment. The changing physical capacity between assessments ensures that soldiers develop and demonstrate the physical qualities required for their specific occupational demands. A deep understanding of which physical qualities are required for soldiers to complete numerous high-demanding combat maneuvers can expedite recruitment processes, ensure readiness, and help target physical training interventions.

Soldiers undergo a multitude of specific physical and tactical training exercises to ensure they have the required physical qualities for optimal combat performance. General physical qualities include body composition, anthropometrics, and physical fitness metrics such as strength, power, endurance, and mobility. The literature demonstrates that soldiers with greater strength, power, and lean muscle mass than comparable counterparts can move better when carrying/wearing an external load (3,5,9). In occupational and sporting environments, similar metrics have been used to predict performance outcomes and risk of injury (12–17). The ability to predict performance outcomes, particularly in a military setting, may provide insights into a soldier's ability to maneuver efficiently under load in high-risk environments.

Accurate predictive models can offer valuable analytics to enhance combat maneuvers like bounds; however, appropriate feature and model techniques must be utilized to achieve accurate predictions. In the exercise and sports science field, numerous studies have utilized dimension reduction techniques to reduce variable collinearity and the sample size to the number of features ratio (18–22). Integrating feature reduction techniques, such as exploratory factor analysis (EFA), with predictive models has proven to be an effective framework that improves model prediction accuracy in health and performance settings (22–26). EFAs can uncover latent constructs in physical performance datasets (e.g., strength, power, and mobility) via oblique rotations, allowing irrelevant data to be dropped and the remaining variables to identify factors based on correlations or covariances (22,25). Oblique rotations allow for correlated factors, such as multiple factors containing strength and power measures (27); however, they remain independent, representing their own specific qualities (22). Extracted factors can then be utilized in prediction models to determine an individual's combat maneuverability ability.

Predicting outcomes requires models such as logistic regressions (LR), decision-based trees (i.e., random forest (RF)), and neural networks (i.e., multilayer perceptron (MLP)) to identify trends and patterns within the input metric dataset based on the outcome variable (28). However, these models have different architectures, with methodologies that range from linear (i.e., LR) to nonlinear techniques (i.e., RF and MLP) to fit models optimally. Due to the complexity of MLP models and the various hyperparameter tuning methods, they often demonstrate a high accuracy in classifying outcomes compared with other models (29). Classification models have yet to be studied as a means of identifying high and low performers in combat maneuverability; however, they demonstrate the capability to classify high-performing personnel, who consistently complete combat bounds within specific time thresholds, and low performers, who fail to meet these standards (5). In addition, classification models can provide valuable insights into the essential physical qualities required for optimal combat movement and serve as a beneficial tool for identifying high or low combat movement performers.

This study aimed to identify the key physical qualities distinguishing high performers from low performers through an EFA. Factors extracted from the EFA were inputted into suitable classification models to assess their accuracy in classifying participants' performance groups and to determine which factors were most influential in deciphering between high and low performers. A successfully developed model provides proof of concept and may provide military organizations with a purposeful tool to assist in personnel selection for combat operations and optimize physical training interventions for occupational readiness. Two hypotheses were formed for this study: 1) a neural network model will have the highest training and testing accuracy across all classifier models developed due to the models' ability to handle complex data patterns, and 2) a combination of strength and power qualities will be the prominent physical qualities compared with other physical fitness qualities like mobility to distinguish between high and low performers of the fire and movement assessment.

METHODS

Participants. Thirty-four recreationally active adult civilian participants (sex: 9 females and 25 males; age: 28.7 ± 8.9 yr; height: 1.73 ± 0.08 m; mass: 73.0 ± 10.73 kg) gave written informed consent and completed the study protocol. Participants completed the Exercise Sport Science Australian prescreening tool (Supplemental Digital Content, http://links.lww.com/MSS/D279) (30) and were required to be free from medical complications, injuries, and mobility limitations for 6 months before participation. This study was approved by the Macquarie University Ethics Committee (Reference Number: 520231297449904).

Procedures. A cross-sectional study was conducted to evaluate the physical qualities of participants and then develop classification models to predict high and low performers of a continuous high-intensity simulated fire and movement assessment. Participants attended two sessions in a fixed

TABLE 1. The physical tests and assessments conducted in sessions 1 and 2 are displayed in the order in which they were conducted.

Session 1 Tests	Device	Condition	Protocol	Sets/Reps	Rest	Metrics
Shoulder mobility	HumanTrak (VALD, Australia)	NBA and BA	Manufacturers	2/3	1–2 min	Peak sagittal and frontal plane angles
Overhead squat	HumanTrak	NBA and BA	Manufacturers	2/3	1–2 min	Peak flexion of the knee and hip
Countermovement Jump	ForceDecks (VALD, Australia)	NBA and BA	Manufacturers	2/3	3–5 min	Peak jump height
Squat jump	ForceDecks	NBA and BA	Manufacturers	2/3	3–5 min	Peak jump height
Isometric mid-thigh pull	ForceDecks	NBA	Manufacturers	3/1	3–5 min	Kinetics (i.e., peak forces and RFD)
Isometric grip strength	Baseline Dynamometer	NBA	Hamilton et al. (31)	3/1	3–5 min	Kilograms
Isometric hip abduction and adduction	ForceFrame (VALD, Australia)	NBA	Manufacturers	1/3	N/A	Kinetics (peak forces)
Isometric push-up	ForceDecks	NBA	Bellar et al. (32)	3/1	3–5 min	Kinetics (i.e., peak forces and RFD)
Multistage fitness test	Official Audio and Polar Heart Rate Monitor	N/A	Ramsbottom et al. (33)	N/A	N/A	$\dot{V}O_{2max}$ and peak HR, recovery HR (60 s)

Session 2 Tests	Device	Condition	Protocols	Sets/Reps	Rest	Metrics
Drop landings (40 cm)	ForceDecks	NBA and BA	Manufacturers	2/3	3–5 min	Drop landing force
Fire and movement	ADF Audio	BA	Silk and Billing (7)	N/A		Laps ("bounds") achieved

Rest was shown as the break between sets. ADF, Australian Defence Force; NBA, no body armor; BA, body armor (23.5 kg); RFD, rate of force development; O_2 max, maximal oxygen uptake; HR, heart rate.

sequence, with at least 3-d rest between each session. Sessions were conducted in an indoor gymnasium. Session 1 involved a physical test battery, where loaded and nonloaded conditions were used throughout (Table 1). Given the multifactorial demands of military tasks (34,35), assessments of strength, power, aerobic endurance, mobility, and body composition were conducted to ensure that all relevant physical qualities were measured. The body armor (BA) loaded condition replicated the ADF combat load of 23.5 kg, whereas the nonloaded condition required no body armor (NBA) to be worn (4). A weighted vest simulated BA and included 4.5-kg metal plates for the front and back, 1.25-kg metal blocks, and rags arranged to evenly distribute the mass and reduce asymmetrical load disturbances. Session 2 involved 40-cm drop landings (both conditions) and the fire and movement assessment (loaded condition) (7). As depicted in Table 1, a logical sequence of the tests was completed for each session to decrease injury risk. Standardized instructions, encouragement, and demonstrations were provided. Maximal effort on all physical tests and assessments was required.

Participants' mass, height, and body composition measures were collected before the commencement of session 1. Only mass was collected upon participants' arrival in session 2. Table 1 displays sessions 1 and 2 in the order of the tests performed, protocols followed, and a description of the metrics collected. Body composition analyses were completed using the Styku (Model S100X; Styku, Los Angeles, CA); protocols were followed per the manufacturer's recommendations. Physical tests that utilized the VALD systems followed the manufacturer's protocol and can be found here (36). The grip strength test followed the protocols established by Hamilton et al. (31), the isometric push-ups followed the protocols by Bellar et al. (32), and the multistage fitness test protocol by Ramsbottom et al. (33). The cutoff threshold for the fire and movement assessment was 50 laps ("bounds"), approximately four times greater than the minimum standard of the ADF All Corps. (10). Fifty bounds were set as a logical estimate that most people would fail within this criterion. Failure to complete a bound was determined by a participant's inability to traverse 6 m in 5 s, moving from a starting prone position to a kneeling rifle position at the 6-m mark. One successful bound was defined as the participant completing a 6-m traverse in less than or equal to 5 s. Figure 1 illustrates the fire and movement protocol adapted to a lab setting (7). The minimum fire and movement assessment standards were not disclosed to encourage maximal effort. This testing was part of a large study investigating performance prediction in various military performance tests, including the lift-to-place and jerry-can-carry assessment. Those tests were not included in the present analysis and have been described previously (37). Additional test protocol information is provided in the Supplemental Digital Content (http://links.lww.com/MSS/D279).

Instrumentation, validity, and data processing. Kinematics during the shoulder and overhead squat mobility tasks

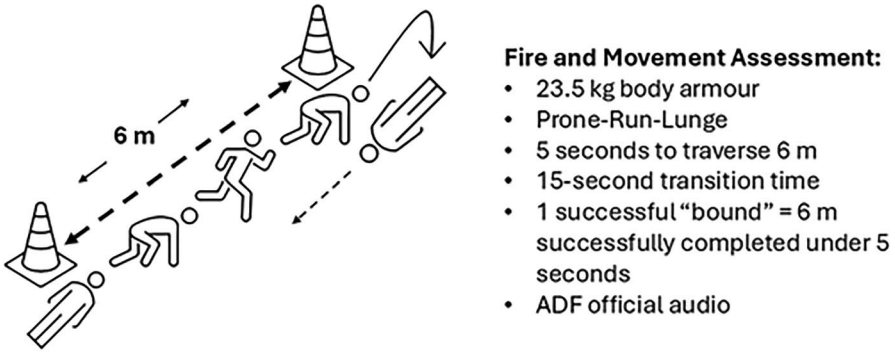


FIGURE 1—Silk and Billing (7) Australian Defence Force (ADF) fire and movement protocol adapted into a laboratory-based setting.

for unloaded and loaded conditions were captured by a validated markerless motion capture system at 30 Hz (HumanTrak, VALD, Australia, V 2.9.3; standard error of measurement = 3.24°–5.29°) (38). A custom MATLAB (R2021a, MathWorks) script filtered the raw kinematic data via a dual-pass fourth-order Butterworth filter at 8 Hz and extracted peak angles at the hip and knee from the overhead squat tests and peak sagittal and frontal plane shoulder angles. The isometric mid-thigh pull (IMTP), isometric push-up, countermovement jump (CMJ), squat jump (SJ), and drop landing (40 cm) data were collected by VALD's ForceDecks (VALD, Australia, V 2.9.3; 1000 Hz). VALD's proprietary algorithms calculated all kinetic metrics: peak force, force at specified times, rate of force development (RFD), and jump height. For the CMJ, the impulse-momentum equation was used for the jump height calculation, whereas for the SJ, the flight time equation was calculated for the jump height metric (39,40). Participants' best results from the appropriate tests were extracted from VALD Hub (a single online centralized platform). The VALD ForceDecks automated metrics were left unfiltered due to the strong validity findings of VALD's derived performance metrics (41,42). Data from the VALD ForceFrame (VALD, Australia, V. 3.4.7; 50 Hz) were unfiltered as per previous reliability studies (43,44): proprietary algorithms calculated peak force.

Circumferences and lean muscle mass were estimated by the Styku system's proprietary machine vision algorithms. The Baseline Hydraulic Hand Grip Dynamometer (Fabrication Enterprises, White Plains, NY) measured participants' grip strength: the best repetition from the participants' dominant limb in kilograms was manually recorded. Estimated maximal oxygen uptake ($\dot{V}O_{2max}$) was calculated based on the participant's multistage fitness score (33): heart rate (HR) data were recorded continuously using a chest-strap monitor (H10; Polar Electro Oy, Kempele, Finland) to obtain peak HR and post–60-s HR recovery. Upon participant completion of the fire and movement assessment, the final bound achieved and the rate of perceived exertion scores (6–20) were manually recorded (45).

Data cleaning, statistical analysis, and machine learning. Python's Jupyter Lab (V. 4.0.11) was used for all data cleaning and predictive model development. A complete list of all Python packages and their versions required for model development can be found in the Supplemental Digital Content (http://links.lww.com/MSS/D279). Participants' fire and movement assessment results were categorized into two groups: a high performer group, where participants reached the maximum threshold of 50 bounds, and a low performer group, where participants failed to reach the maximum threshold (less than 50 bounds). Descriptive statistics of each group's fire and movement assessment results were calculated and then removed from the participants' dataset.

Exploratory factor analysis. Factor extraction and statistical analyses were performed using IBM SPSS Statistics Version 29 (IBM Corp, Armonk, NY). Numerous studies in the exercise and sports science field have utilized dimension reduction techniques with a similar sample size that is presented in this study (18–20). An EFA was conducted to concisely describe and interpret participants' physical performance measures. The remaining metrics were standardized to a zero mean and a standard deviation of one in preparation for the EFA. A correlation matrix, Bartlett's test of sphericity, and a Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy were conducted to ensure the performance metrics met the required assumptions to complete an EFA. Criteria for determining factor adequacy were established *a priori*. Scree plot, interpretability, and simple structure were assessed to decide the optimal number of factors. Due to the nature of the metrics, it was assumed that factors would be correlated, so a Promax rotation was employed. Pattern coefficients ≥ 0.3 were set. Factor values were taken alongside their categorized high or low performer label, forming the dataset for the predictive models. Independent-samples two-tailed *t*-tests and Cohen's *d* effect sizes (ES) were also performed on each factor to analyze differences between performance groups. If specific metrics were deemed relevant to the fire and movement assessment that did not meet the EFA criteria, an independent *t*-test was run on these specific metrics (i.e., $\dot{V}O_{2max}$, based on Foulis et al. [8] findings). Assumptions of normality were met prior to conducting *t*-tests, except for factor 4, which failed the Levene's test ($P = 0.034$). Thus, a Welch's *t*-test was used for factor 4's analyses. Statistical significance was set *a priori* at $\alpha = 0.05$. Cohen's *d* ES was calculated. ESs were classified as trivial ($d \leq 0.2$), small ($d > 0.2$), moderate ($d > 0.5$), large ($d > 0.8$), and very large ($d > 1.3$) (46). Eigenvalues, the total variance explained, and interfactor correlations were also extracted from SPSS.

Training phase. All classification model Python scripts can be accessed via GitHub (47). The predictive model dataset was randomly split into 80% training (14 low performers and 13 high performers) and 20% testing (4 low performers and 3 high performers) datasets (48). Three classification predictive models were selected as appropriate models to classify high or low performers for the fire and movement assessment: LR, MLP, and RF models (12,49,50). The EFA extracted factors were labeled as the predictor metrics, and high versus low performer classification was the outcome metric. All machine learning models were optimized through a randomized grid search (*scikit-learn 1.4.2*, *RandomizedSearchCV*) with 2000 iterations, using a leave-one-out cross-validation and classification accuracy to identify the best parameters from the training dataset. Hyperparameter distribution ranges for each model can be viewed in the Supplemental Digital Content (http://links.lww.com/MSS/D279). Once the hyperparameters were optimized, they were fitted to each respective model. The fitted models were then used to predict the performance group for each participant in the training set. A confusion matrix chart, alongside accuracy, recall, and precision values, were calculated to assess each model's training performance.

Testing phase. An accuracy, recall, and precision of greater than or equal to 80% were required results for each model during the training phase to assess their classification performance on the unseen testing dataset. Models that achieved this threshold were introduced to the testing dataset,

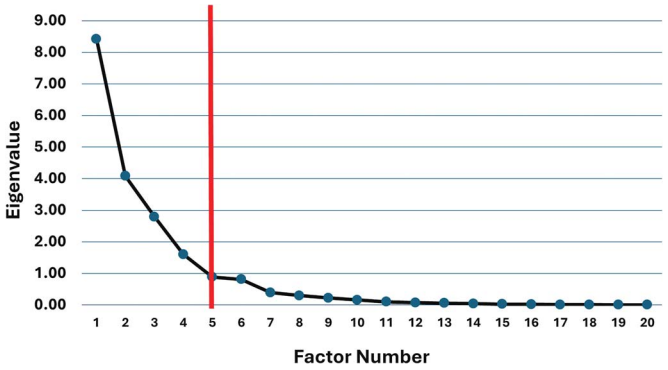


FIGURE 2—The Exploratory Factor Analysis (EFA) scree plot demonstrates the "elbow flexion" point, allowing the optimal number of four factors to be chosen.

where participants were predicted as either high or low performers based on their EFA factor values. Again, a confusion matrix chart and accuracy, recall, and precision values were formed. In addition, factor importance plots specific to each model's architecture were formed to explain how each factor influenced the model's ability to predict if a participant was a high or low performer. For the LR, coefficient values; for the MLP, permutation importance values; and for the RF, feature importance values were taken, respectively, and plotted in ascending order.

RESULTS

The fire and movement assessment resulted in 16 participants being labeled as high performers, completing 50 bounds with an average RPE of 16.3 ± 1.6, and 18 participants labeled as low performers, completing less than or equal to 40 bounds (18.94 ± 14.57) with an average RPE of 17.2 ± 2.0. Only males formed the high performer group (assessment score: 50; age: 25.8 ± 6.4 yr; height: 1.76 ± 0.08 m; mass: 74.2 ± 8.23 kg), whereas the low performer group observed a bimodal distribution of 40 or less bounds and had an even split of nine males (assessment score: 28.44 ± 10.60; age: 27.0 ± 10.0 yr; height: 1.76 ± 0.05 m; mass: 79.8 ± 9.98 kg) and nine females (assessment score: 9.44 ± 11.65 age: 35.7 ± 10.3 yr; height: 1.65 ± 0.07 m; mass: 64.1 ± 10.30 kg). All metrics from the multistage fitness, body composition, isometric hip abduction and adduction, shoulder mobility, and NBA drop landing tests were removed as they did not meet the input criteria for the EFA. The criteria not met were low correlations demonstrated in the correlation matrix or negatively impacted the KMO result. The accepted EFA metrics resulted in an acceptable Bartlett's test of sphericity result ($p \leq 0.001$; $\chi^2 = 1040.73$) and a KMO value of 0.675, meeting the assumptions that an EFA could reduce the data dimensions. The interpretability criterion (simple structure) and examination of the scree plot (Fig. 2) suggested that four factors were required. Four factors resulted in 81.46% of the variance explained. Factor coefficient loadings and communalities values from the rotated Promax solution are presented in Table 2; interfactor correlations were deemed trivial to moderate (51) and can be seen in the Supplemental Digital Content (http://links.lww.com/MSS/D279).

TABLE 2. Rotated pattern matrix loading coefficients, extracted communalities, and the metrics used within the Exploratory Factor Analysis (EFA) are displayed.

	Pattern Matrix				
	Factor				
Metrics	1	2	3	4	Communalities
LCMJ jump height (Imp-Mom) (cm)	**1.008**	−0.091	−0.024	0.026	0.929
CMJ jump height (Imp-Mom) (cm)	**0.997**	−0.094	−0.029	0.035	0.908
LSJ jump height (Flight Time) (cm)	**0.948**	0.102	−0.101	−0.033	0.906
SJ jump height (Flight Time) (cm)	**0.913**	0.025	−0.056	−0.028	0.798
Grip strength (kg)	**0.757**	−0.018	0.067	−0.023	0.602
PU peak vertical force (N)	**0.717**	0.117	0.252	−0.051	0.815
IMTP peak vertical force (N)	**0.629**	0.238	0.116	0.046	0.698
BA relative peak drop landing force (kg)	**−0.502**	0.140	0.094	−0.037	0.179
IMTP force at 150 ms (N)	−0.079	**1.034**	0.046	0.023	0.999
IMTP force at 100 ms (N)	−0.036	**1.008**	−0.078	−0.017	0.993
IMTP force at 200 ms (N)	−0.034	**0.921**	0.172	0.073	0.860
IMTP force at 50 ms (N)	0.074	**0.899**	−0.193	−0.076	0.906
PU RFD—100 ms ($N \cdot s^{-1}$)	−0.095	0.001	**1.025**	0.008	0.972
PU RFD—150 ms ($N \cdot s^{-1}$)	0.034	−0.017	**0.978**	0.001	0.990
PU RFD—50 ms ($N \cdot s^{-1}$)	−0.118	0.032	**0.885**	0.010	0.699
PU RFD—200 ms ($N \cdot s^{-1}$)	0.216	−0.085	**0.820**	−0.033	0.861
BA OHS knee flexion avg (°)	0.138	0.002	−0.074	**0.928**	0.917
BA OHS hip flexion avg (°)	0.072	0.011	−0.014	**0.885**	0.820
OHS hip flexion avg (°)	−0.179	−0.008	0.072	**0.866**	0.716
OHS knee flexion avg (°)	0.011	0.004	0.001	**0.846**	0.723

[a]Major loadings are bolded. Extraction method: principal axis factoring. Rotation method: Promax with Kaiser normalization. Rotation converged in six iterations. Communalities values are from the extraction.

Avg, average; CMJ, countermovement jump; LCMJ, loaded countermovement jump; LSJ, loaded squat jump; OHS, overhead squat; PU, push-up; SJ, squat jump; IMTP, isometric mid-thigh pull; BA, body armor (23.5 kg); RFD, rate of force development.

TABLE 3. Independent *t*-test values of each factor and estimated $\dot{V}O_{2max}$ scores from the EFA.

Factor	High Mean	Low Mean	*t* Statistic	*p* Value	ES (*d*)
Factor 1: strength, jump, and landing	0.78 ± 0.54	−0.69 ± 0.79	6.40	<0.001	2.15
Factor 2: IMTP force (50–200 ms)	0.40 ± 1.13	−0.35 ± 0.82	2.21	0.036	0.77
Factor 3: PU RFD	0.44 ± 1.12	−0.39 ± 0.94	2.35	0.026	0.81
Factor 4: OHS ability[a]	0.36 ± 0.58	−0.32 ± 1.17	2.16	0.040	0.73
Estimated $\dot{V}O_{2max}$	45.90 ± 7.15	37.00 ± 9.21	3.13	0.004	1.08

[a]Welch's *t*-test.
ES, effect size (Cohen's *d*); High, high-performance group; Low, low-performance group; IMTP, isometric mid-thigh pull; OHS, overhead squat; PU, push-up; RFD, rate of force development.

All factor coefficients are represented in a positive direction, besides BA peak drop landing force relative to body mass (newtons per body mass (kilograms)), represented in factor 1, which is in the negative direction. Factor 1, accounting for 41.28% of the total variance explained, represents total body strength, maximum jump height ability in loaded and nonloaded conditions, and the ability to dissipate force when landing from a box height of 40 cm. Factor 2, accounting for 19.95% of the variance explained, represents the force produced at 50–200 ms during the IMTP. Factor 3, accounting for 13.04% of the variance explained, represents the RFD from the isometric push-up test. Factor 4 (7.19% total variance explained) represents the ability to complete an overhead squat for depth with and without BA.

Table 3 compares the factors and estimated $\dot{V}O_{2max}$ scores between high and low performers. All factors and estimated $\dot{V}O_{2max}$ demonstrated a significant difference ($p < 0.05$). Factor 1 had a very large ES, whereas factors 2–4 and estimated $\dot{V}O_{2max}$ were deemed moderate–large. Notably, all the mean factor values for the high performer group were positive, whereas the mean values for the low performers were negative.

For all machine learning models developed, training accuracy, precision, and recall were all greater than or equal to 80% for predicting high and low performers in the fire and movement assessment. All models demonstrated a high accuracy in the training phase, with an 85%–88% success rate. Training phase confusion matrices, results, and hyperparameters can be seen in the Supplemental Digital Content (http://links.lww.com/MSS/D279). Testing phase confusion matrices revealed that only the LR model had a 100% classification accuracy. The RF and MLP models had an accuracy of 86%, where they both mislabeled the same high-performing participant incorrectly, likely due to a low factor 1 score. The results from the testing phase, along with each participant's respective factor values, are presented in the Supplemental Digital Content (http://links.lww.com/MSS/D279). Figures 3–5 display the influence of each factor's predictability for each model. For all models, factor 1 had the largest influence (approximately six times greater influence than factors 2–4) on predicting the performance group of an individual. Factor 4 had a minor influence across all models. Factors 2 and 3 showed zero to minimal influence, dependent upon the model.

DISCUSSION

This study aimed to group physical qualities via an EFA and then utilize the optimal factors extracted to train machine learning models to classify high (50 bounds) and low performers (less than 50 bounds) for a fire and movement assessment. The EFA produced four optimal factors with significant differences detected across all factors between high and low performers (Table 3). The developed machine learning classifier models made high and low performance group predictions based on specific physical qualities represented by the four factors; the LR model correctly classified all participant's performance groups (100% accuracy) in the testing set, whereas one high performer participant was mislabeled by both the MLP and RF models (86% testing phase accuracy). Hypothesis 1 proclaimed that the MLP classifier model would yield the highest testing accuracy; our results demonstrated that there was only one misclassification error difference between the three models. This hypothesis was rejected due to similar model classification results in the testing phase, demonstrating proof of concept that accurate predictive models can be developed. Factor 1, representing strength, maximal jump height, and the ability to dissipate force during drop landings, was the most influential factor in predicting a high or low performer in the fire and movement assessment, with an approximate influence of six times greater than the other factors (Figs. 3–5). This led to the acceptance of our second hypothesis, as strength and power qualities were hypothesized to have the best prediction qualities. Our results indicate that physical preparation that emphasizes lower- and upper-body strength

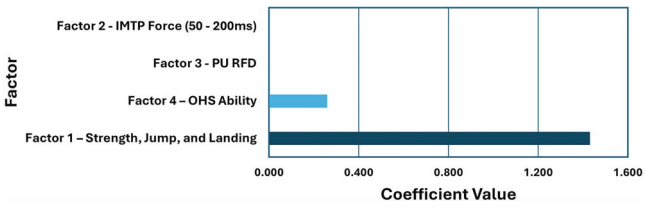


FIGURE 3—Factor importance based on coefficients from the logistic regression classification model. The isometric mid-thigh (IMTP) force developed at 50–200 ms is represented in factor 2. The push-up rate of force development (PU RFD) is represented in factor 3. Overhead squat (OHS) ability is represented in factor 4.

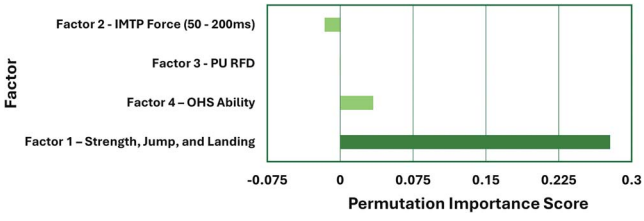


FIGURE 4—Permutation importance of the multilayer perceptron classification model. The isometric mid-thigh (IMTP) force developed at 50–200 ms is represented in factor 2. The push-up rate of force development (PU RFD) is represented in factor 3. Overhead squat (OHS) ability is represented in factor 4.

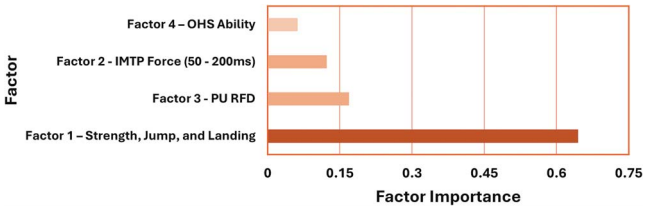


FIGURE 5—Factor importance of the random forest classification model. The isometric mid-thigh (IMTP) force developed at 50–200 ms is represented in factor 2. The push-up rate of force development (PU RFD) is represented in factor 3. Overhead squat (OHS) ability is represented in factor 4.

and power enhances military personnel's performance in combat movement scenarios. Thus, military organizations should promote physical assessments that monitor IMTP, isometric push-up peak force, jump height in the CMJ and SJ (both conditions), grip strength, and drop landing abilities, as our preliminary developed models had high accuracy predicting high and low performers of the fire and movement assessment.

Numerous exercise and sports science studies have employed data dimension reduction techniques to break down physical performance metrics into factors/components for visualizations, interpretations, and further analyses (19,20,52). Data dimensions were reduced via an EFA to group predictor metrics collected from the physical test battery into factors that express strong relationships. Four factors were formed, each uniquely representing specific physical qualities (Table 2). Factor 1 explained the most variance (41.28%) in the dataset. The remaining factors explain a small but relevant portion of the variance, providing additional and unique factors for future analyses. These four factors provide a general representation of physical quality domains required for optimal health and performance, although aerobic qualities were not included in the EFA analysis, as the aerobic metrics obtained did not meet the EFA input criteria (53). Our results demonstrate that EFAs can be a useful tool in physical characterization profiling, with four specific yet unique factors extracted in this study. Understanding how metrics cluster together helps provide insight into which physical qualities may be leveraged or improved to enhance combat maneuverability.

Predictive models within military and sports settings have seen notable progress in model development and implementation within recent years (12,15,16). The LR model achieved perfect testing phase accuracy, whereas the MLP and RF models had one misclassification error (86% testing phase accuracy). Our developed models have demonstrated the potential for accurate predictions within a military environment for a specific combat assessment across three architecturally different models. A methodology review across 72 studies investigating LR and artificial neural network models found that generally, LR and neural networks like MLP models outperform models like RF models that utilize a decision tree structure (29). This contradicts our current findings, as all models performed similarly across the training and testing phases. This outcome could be due to a relatively simple model feature (metric) structure (i.e., using only four factors to train and classify outcomes), as complex models like the MLP benefit from increased sample sizes and further feature input to extract and identify patterns, to optimize model fit (54). Furthermore, our models show comparative performance to Ab Rasid et al. (55), who developed an LR model to predict skateboard potential from 12 fitness and neuromuscular input metrics. Only two misclassifications of high and low potential skateboarders were made from a testing sample size of 13. More applicable to our research is a study on injury risk in special forces operators using a structured decision-tree model, which successfully utilized physical metrics like strength, power, body composition, and kinematic asymmetries to predict an operator's likelihood of an injury to an accuracy of 87% (16). Supporting the use of predictive models trained on physical quality input metrics in a military environment. Optimizing our predictive models with larger datasets is expected to increase each model's generalizability and reduce the risk of overfitting as models are then exposed to increased diversity and variations within the training datasets (i.e., military personnel and high-performing female datasets) compared with the current training set (56). All model types are expected to be suitable for classifying fire and movement performance outcomes. This instills optimism about future model implementation in a military setting, which may lead to targeted selection, training, and performance improvement strategies.

The ability to optimize combat movement performance through targeted physical interventions is a crucial criterion for military organizations in developing current and incoming combat soldiers. As all factors differed significantly between groups (Table 3), factor importance within the predictive models was extracted to highlight the most influential factors to classify a high and a low performer in the fire and movement assessment. Across all models, the number one ranked predictor was factor 1. An observable higher level of influence was exhibited in factor 1 compared with all other factors. This phenomenon is further demonstrated by factor 1's very large ES (2.15) compared with factors 2–4, which have moderate–large ESs (0.72–0.81). Hence, to optimize combat maneuverability, interventions should enhance mechanics and performance for maximal jump height in the CMJ and SJ (NBA and BA), strength in the sagittal plane of the hip and knee, horizontal upper body push strength, grip strength, and drop landing ability. Although all factors differed significantly, the aforementioned movements and exercises should be prioritized to increase fire and movement performance over the other physical qualities represented in factors 2, 3, and 4. As there is a crossover between the physical tests and factors, the metrics derived from IMTP and isometric push-ups found across factors 1–3 will be concurrently trained as factor 1's physical qualities are enhanced.

Our findings match the broader military physical performance literature, demonstrating that optimal physical qualities, such as strength, power, and mobility, as represented in our factors, lead to enhanced military task performance (57–60). Our results add to the findings of Stein et al. (5), who found the standing broad jump and running critical velocity predicted 74% of the fire and movement assessment variance by Silk and Billing (7). Foulis et al. (8) also revealed that the multistage fitness test was predictive of combat movement, which matched our findings that superior $\dot{V}O_{2max}$ is a quality

of a high performer in the fire and movement assessment. It has been shown that soldiers' vulnerability to enemy fire rises with each subsequent combat bound, demanding a continued high-intensity effort and cardiovascular endurance for each combat movement (2,8,61). Although repeated anaerobic capacity, like sprint efforts, was not tested, and aerobic endurance metrics during the EFA were removed, our results suggest that assessing these qualities may be unnecessary because of our model's strong testing phase results. Overall, military organizations should target improving all physical fitness and health factors, with an emphasis on strength, power, and drop landing performance.

This research demonstrated limitations. Throughout both testing sessions, additional physical tests occurred. Reasons for exclusion were that predictor metrics did not meet the criteria for an EFA or if physical assessments were specifically completed for another research question. Fatigue may affect test outcomes; however, 3- to 5-min rest periods were provided in each session (62). The EFA's correlation matrix used a small sample (34 participants), prone to sampling error. Research shows that eigenvalues and factor loadings can differ with larger samples in EFA analysis (63). However, evidence suggests that EFA can yield stable solutions even with less than five participants per variable, particularly under conditions of high factor loading values and minimal extracted factors (63), a result our study exhibits alongside acceptable KMO and Bartlett's sphericity results. The fire and movement assessment results showed a sex imbalance: a male-only high-performing group, whereas both sexes formed the low-performing group. Physical differences between the sexes, such as greater strength in males, may lead to models that overfit these traits for high-performance classification and *t*-test results that reveal significant factor differences reflecting these biological disparities (64). Lastly, our sample size was relatively small, reflecting only a small portion of the general population; increased sample size to include military personnel and high-performing females will improve generalizability and the validity of our developed models. Additional training data can reduce the risk of overfitting and the statistical likelihood of misleading classification results, which may be observed in a small test size such as ours (i.e., seven participants).

CONCLUSIONS

High-speed, efficient movement under enemy fire is a multidimensional physical activity. Our developed machine learning models (LR, MLP, and RF) demonstrated proof of concept with accurate results in classifying high (50 bounds) and low performers (less than 50 bounds) for the fire and movement assessment. Military organizations can utilize these preliminary prediction models to monitor combat performance, assess readiness, enhance recruitment strategies, and leverage the physical characteristics observed in factor 1 to design physical interventions to improve overall survivability. Factor 1 physical qualities that separate high and low performers in the fire and movement assessment are peak jump height (CMJ and SJ—NBA and BA), peak force in the IMTP and isometric push-up, maximum kilograms in the grip strength test, and reducing peak landing force relative to body mass from a 40-cm platform. Individuals who possessed either high or low factor scores were generally correctly classified. High performers trended with higher factor scores, and low performers exhibited lower factor scores. Although our models demonstrated accurate classification results, it is essential to note that they were trained on data from a small general population sample. An increased sample size and variation in populations, including military personnel and high-performing females, compared with our current study, can enhance the EFA stability and optimize the generalizability and fit of our predictive models.

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